Decay-Aware Error Budgets for Recurrent-State SVD

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Abstract

SinkState stores linear-attention recurrent states in adaptive low-rank form and truncates components with a relative singular-value threshold. Such a threshold does not directly bound state or readout error accumulated across updates. We extend the scalar-decay recurrence with a certified Frobenius-error budget: if the current error bound is b_{t-1}, decay is $\alpha_t \in [0,1]$, and truncation residual is at most δ_t, then $b_t = \alpha_t b_{t-1} + \delta_t$ is a bound. Choosing a rank with $\delta_t \le B - \alpha_t b_{t-1}$ preserves a fixed budget B and bounds each readout error by $B\|q_t\|_2$. An exact-rational CPU experiment checks 5,120 synthetic diagonal-state steps with zero violations. The experiment does not validate model quality, GPU speed, or the exact-truncation premise for SinkState's local/frozen update.

1 Introduction

Linear-attention systems update a recurrent matrix state and read it with the current query. SinkState observes low effective rank in such states and maintains adaptive low-rank factors using buffered updates, local SVD, and asynchronous consolidation (Jiang et al., 2026). Its design truncates components relative to the largest singular value. A relative threshold is a rank rule, but does not itself state a cumulative error budget for a long sequence of updates.

We derive a simple state-error certificate for scalar decay and exact low-rank truncation, then test it using exact rational arithmetic on a deliberately restricted diagonal recurrence. The extension exposes a trade-off: a guaranteed error budget may retain more components than a relative threshold. It does not show that the guarantee can be added to SinkState's local/frozen algorithm without further residual accounting.

The source arXiv:2611.14246v1 first appears in the 2026-11-11 daily feed and is carried into the November 12 snapshot. Its identifier and title were absent from prior paper artifacts and paper-specific Git history before this work. The feed contained title/abstract metadata; the full v1 paper was consulted. This audit covers repository history and available session context, not external access logs.

2 Decay-aware truncation certificate

Let the exact state follow

$$H_t = \alpha_t H_{t-1} + u_t v_t^\top, \qquad 0 \le \alpha_t \le 1, \quad (1)$$

and let the compressed state start from $\widehat{H}_{t-1}$. Before truncation, apply the same decay and rank-one update to obtain $\widetilde{H}_t = \alpha_t \widehat{H}_{t-1} + u_t v_t^\top$. Thus

$$\|H_t - \widetilde{H}_t\|_F = \alpha_t \|H_{t-1} - \widehat{H}_{t-1}\|_F. \quad (2)$$

Let the truncation be $\widehat{H}_t = \widetilde{H}_t - R_t$, with $\|R_t\|_F \le \delta_t$. The triangle inequality yields

$$\|H_t - \widehat{H}_t\|_F \le \alpha_t \|H_{t-1} - \widehat{H}_{t-1}\|_F + \delta_t. \quad (3)$$

If b_0 bounds initial error and $b_t = \alpha_t b_{t-1} + \delta_t$, induction gives $\|H_t - \widehat{H}_t\|_F \le b_t$. For a fixed target B, it suffices to choose truncation residual

$$\delta_t \le B - \alpha_t b_{t-1}. \quad (4)$$

For an exact SVD of $\widetilde{H}_t$, rank selection uses the tail certificate $\sum_{i>r} \sigma_i^2 \le (B - \alpha_t b_{t-1})^2$. For any readout query q_t,

$$\|(H_t - \widehat{H}_t) q_t\|_2 \le \|H_t - \widehat{H}_t\|_F \|q_t\|_2 \le b_t \|q_t\|_2. \quad (5)$$

This is a state and readout bound, not an output-quality guarantee for a nonlinear model.

The certificate requires scalar decay, identical updates to the exact and approximate states, and a valid upper bound on the residual created by truncation. `DecayBudgetClaims.lean` checks the scalar one-step and sequence invariant with Lean

Measure	Relative	Budget
Mean retained rank	10.79	12.28
Mean trial max readout error	3.428	0.116
Maximum observed proposed-policy error	–	0.130
Checked steps	–	5,120
Certificate violations	–	0

Table 1: Exact-rational diagonal surrogate.

4.35.0 and Mathlib 4.35.0. It does not formalize matrix norms, the SVD tail-residual identity, or an implementation. SinkState's local top-subspace update and frozen factors do not necessarily equal exact global truncation. Applying Equation 3 to that implementation requires additional certificates for local update error and frozen-tail effects.

3 Exact-rational CPU experiment

The standard-library script uses 32 trials of 160 steps at width 16, seed 20261114. States are diagonal; each update is a rank-one write on a single diagonal coordinate. Decays alternate between $3/4$ and $9/10$, and writes are seeded signed rationals. Thus singular values are exact absolute diagonal entries, and all truncation and norm comparisons use `fractions.Fraction`. The relative-threshold baseline drops values below 0.1 of the largest. The proposed policy uses $B = 1/2$ and drops the largest tail allowed by $B(1-\alpha_t)$, which is the fixed-budget rule when the previous certificate is B.

For diagonal error matrices, the maximum coordinate error equals the spectral norm and is bounded by the Frobenius norm. The script checks the actual exact state error against B at every proposed-policy step. The relative-threshold baseline is not constrained by this budget. Results are deterministic and reproducible by running `python3 experiment.py`.

4 Limitations

This experiment is not a general SVD, a model inference run, or a GPU benchmark. The diagonal state makes spectral truncation exact and omits randomized SVD approximation, floating-point roundoff, buffered updates, local/frozen-factor drift, factor storage, and compaction cost. Lean checks the scalar recurrence invariant, not the matrix/SVD premises or paper experiment. The available hardware is an Apple M2 without NVIDIA GPU or CUDA. A next study should measure certified residuals inside SinkState's actual local/frozen update and test accuracy, rank, memory, and latency on its recurrent model checkpoints.

Reproducibility

Run `python3 experiment.py` in this directory. It writes `results.json`. From `papers/formal`, run `lake env lean DecayBudgetClaims.lean`. The experiment uses only the Python standard library and exact rational arithmetic.

References

Haolin Jiang, Weihang Li, Zhongzhu Zhou, Jinwen Pan, Santosh Pandey, and Hang Liu. 2026. SinkState: On-the-Fly SVD of Recurrent States for Efficient Hybrid LLM Serving. arXiv:2611.14246v1. https://arxiv.org/abs/2611.14246v1.